How Ozone Forms Near Earth's Surface

Use props to engage an audience of learners as you tell an interactive story about the formation of ground-level ozone. This activity is best done along with a visit to your ozone garden, or as an introduction to further explorations about ozone.

Audience: Ages 5 - Adult

Time: 15 minutes

Education Standards

- **K-ESS3-3**. Communicate solutions that will reduce the impact of humans on the land, water, air, and/or other living things in the local environment.
- **5-ESS3-1**. Obtain and combine information about ways individual communities use science ideas to protect the Earth's resources and environment.
- **MS-ESS3-3**. Apply scientific principles to design a method for monitoring human impact on the environment
- **HS-ESS3-4**. Evaluate or refine a technological solution that reduces impacts of human activities on natural systems.
- **SEP6**. Constructing Explanations and Designing Solutions

Learning Goal

- Learners will understand that ground-level ozone forms as the result of other air pollutants combining in the presence of sunlight.

Materials

- Ozone formation props
- Stick-on velcro or tape
- Foam balls in different colors and sizes to represent oxygen, carbon, nitrogen, and solid carbon atoms (optional)
- *What You Can Do to Improve Air Pollution* infographic (optional**)**

Preparation

- Print ozone formation props and cut out the oxygen atoms and ozone label. Laminate them for best results. Consider making mulitple copies of the molecule props so you can engage a larger audience of learners.
- Use stick-on velcro tabs on the molecule props to build ozone during the storytelling. Before beginning the story, add one oxygen atom to the N-O prop to form NO_2. During the story you will remove that oxygen atom from the NO_2 molecule and attach it to the O_2 molecle to make ozone. You will also attach the 'ozone' label to the O_2 molecule prop once ozone is formed.

Directions

- Gather your group of learners in an open space and use the props and script provided to tell the story of ozone formation.
- Ask for volunteers to hold each of the props and pass them out. Tell them that when they hear their part mentioned in the story to hold up their prop and act out what the story says is happening. The props are queued in brackets within the story script.
- Read aloud using the How Ozone Forms script. Cue volunteers when appropriate and ask them to add their own interpretations to create a more engaging experience.

NATIONAL OZONE GARDEN NETWORK

UCAR CENTER FOR SCIENCE EDUCATION

© 2025 University Corporation for Atmospheric Research *All rights reserved*

How Ozone Forms Script

On a warm summer morning in the city, people are waking up and getting ready for their day. The Sun **[Sun]** is already shining brightly.

People use many different types of vehicles to get to where they are going. What kind of vehicles might people use? Cars, buses, motorcycles, scooters, and bicycles... Most of these vehicles **[Factory + Car]** burn gasoline or natural gas, called fossil fuels, to make them go. Nearby factories also burn fossil fuels to make electricity that we use in our homes, schools, and workplaces.

Burning fossil fuels releases invisible gases into the air — carbon dioxide **[CO2]**, nitrogen dioxide **[NO2]**, volatile organic compounds (VOCs) **[VOC]**, and other gases, too. Factories and vehicles **[Factory + Car]** also release solid particles of dust, carbon soot **[C]**, and smoke that are so tiny they can float in the air for a long time.

There are many different gases that make up the air in our atmosphere **[Atmosphere]**. Can anyone think of a type of gas that is in our air? **[O2, CO2]**. The air all around us — which is called the atmosphere **[Atmosphere]** — is mostly made up of nitrogen and oxygen **[O2]** and small amounts of other gases.

Most of the gases we're adding to the air, from things like burning fossil fuels or using industrial chemicals, are considered air pollution. These gases make the air less clean, which is bad for our health. Air pollution can damage our lungs, and harm plants and animals, too.

But on a warm sunny day **[Sun]** when certain types of air pollution are already in the atmosphere, something else can form.

Let's gather the atmospheric "ingredients" needed to make this particular type of air pollution. We need: VOCs **[VOC]**, which can come from car exhaust pipes, or from chemicals in paint or other industrial products; we also need nitrogen oxides like NO2 **[NO2]**, which also come from car exhaust pipes and from factory emissions. Then we add in energy from the Sun **[Sun]**, like on a warm day in the summer. When these ingredients mix with energy from the Sun **[gather these three people/props together and have them "mix" around]**, this starts a chain of chemical reactions that cause an oxygen atom to break off from an NO2 molecule **[detach O from NO2]**. I need one more volunteer to carry this oxygen atom for me **[hand the O to a new volunteer]**.

Now we have a lonely oxygen atom **[O]** floating around in the atmosphere. And when this lonely oxygen atom bumps into an oxygen molecule **[O2],** they stick together to form O3, which is called ozone **[O, O2 — stick on an O and ozone label to become O3 — hand to O volunteer**].

Now way up high in the atmosphere **[Atmosphere]**, ozone **[O3]** is naturally there in the ozone layer. The ozone layer is right here in the stratosphere **[point out the stratosphere in the atmosphere prop]**. Ozone up in the stratosphere is actually really helpful. It helps block UV rays from the Sun that can cause sunburn. But down here, close to the ground, ozone is harmful. It can damage our lungs and make it harder for plants to grow.

As there are more and more cars and factories in our busy cities **[Factory + Car]**, there is a lot more ozone pollution **[O3]**, too. Especially in the summertime when there's more sunlight, ozone levels can be dangerously high. Some people are experiencing health problems because of ozone. To stay safe, it's best to limit outdoor activity on high ozone days.

Thankfully, once ground-level ozone **[O3]** forms it doesn't stay in the atmosphere **[Atmosphere]** for very long, usually only for a few hours, though in really polluted areas, ozone can stick around for several weeks. But anytime the right combination of ingredients are present in the atmosphere **[Atmosphere]** — sunlight **[Sun]**, NO2 **[NO2]**, and VOCs **[VOC]** — there's a good chance that ozone will form. The end!

© 2025 University Corporation for Atmospheric Research *All rights reserved*

Extensions

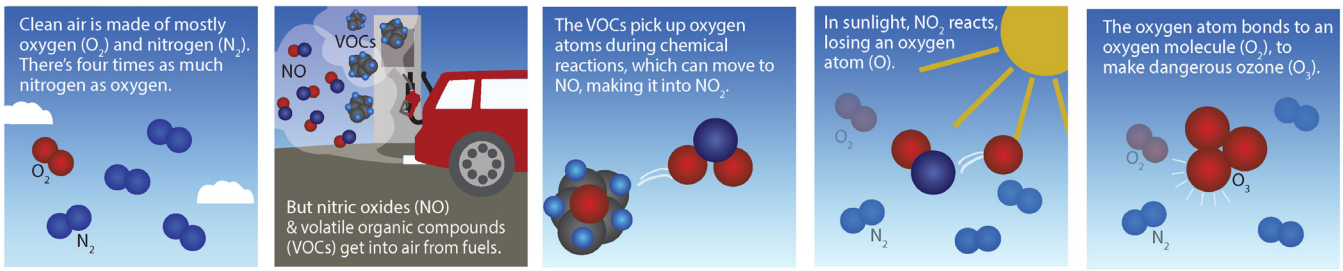
- Ask your audience for ideas about how we could help lower the amount of ozone that forms in the summertime, and ask each person to pledge to tell one person about what they have learned today. Use the *What You Can Do to Improve Air Pollution* infographic to aid in your discussion about solutions
    - Ozone solutions: drive less, carpool, use public transportation, ride your bike, use electric vehicles, fill up gas tanks and use gas-powered lawn equipment later in the day when the Sun isn't out, use VOC-free paints and chemicals, use less electricity, turn off lights and electronics when not in use, use less air conditioning, insulate homes and buildings.
- As a follow up, do the *Putting a Human Face on Ozone Pollution* activity.

Background

In the stratosphere, ozone molecules that make up the ozone layer play an important role — absorbing ultraviolet radiation from the Sun and shielding Earth from dangerous solar energy. But ground-level ozone in the troposphere is both an air pollutant, threatening the health of living things, and a greenhouse gas, trapping heat and contributing to climate change.

A small amount of ozone occurs naturally at ground level, and some ozone molecules migrate down from the stratosphere. But natural sources of ground-level ozone aren't concentrated enough to threaten the health of humans or the environment. Most of the ozone that is found near the ground is formed from chemical reactions with vehicle exhaust and emissions from factories, power plants, and refineries. Since 1900, the amount of ozone near Earth's surface has more than doubled due to the burning of fossil fuels for transportation and industry.

Unlike most other air pollutants, ozone is not directly emitted into the air. Tropospheric ozone is a secondary air pollutant formed by the interaction of sunlight, particularly ultraviolet light, with hydrocarbons and nitrogen oxides, which are emitted by automobile tailpipes and smokestacks. In urban areas, high ozone levels usually occur during warm summer months. Typically, ozone levels reach their peak in mid to late afternoon, after exhaust fumes from morning rush hour have had time to react in sunlight. A hot, sunny day without much air movement is the perfect environment for ozone pollution to form. At the end of the day, as the Sun starts to set, the production of ozone begins to subside.



Ground-level ozone formation. Credit: UCAR

Ozone molecules typically break down quickly, lasting only a few hours or weeks depending on the concentration of pollutants. Ozone concentrations fluctuate greatly with the seasons and with location, with high concentrations of ozone common in urban areas during the warmest months of the year. Local ozone concentration is also influenced by the presence of wildfires or oil and gas refineries. The TEMPO Data Portal shows real-time NO2 concentrations in the U.S. gathered from satellite observations, which indicates where ozone concentrations are currently high. The EPA's AirNow website also monitors air quality, including ground-level ozone concentrations for your local area.

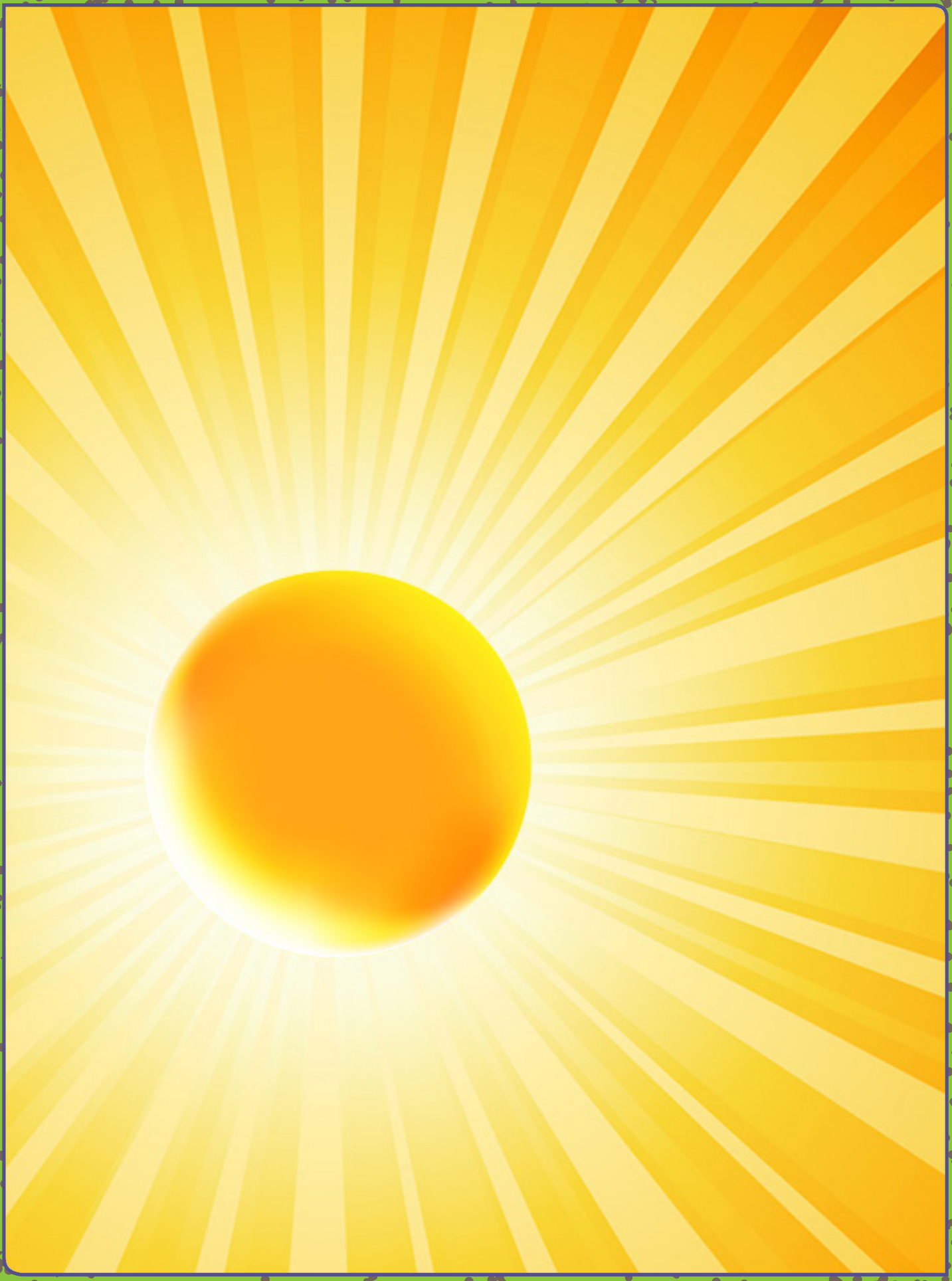
Additional Resources

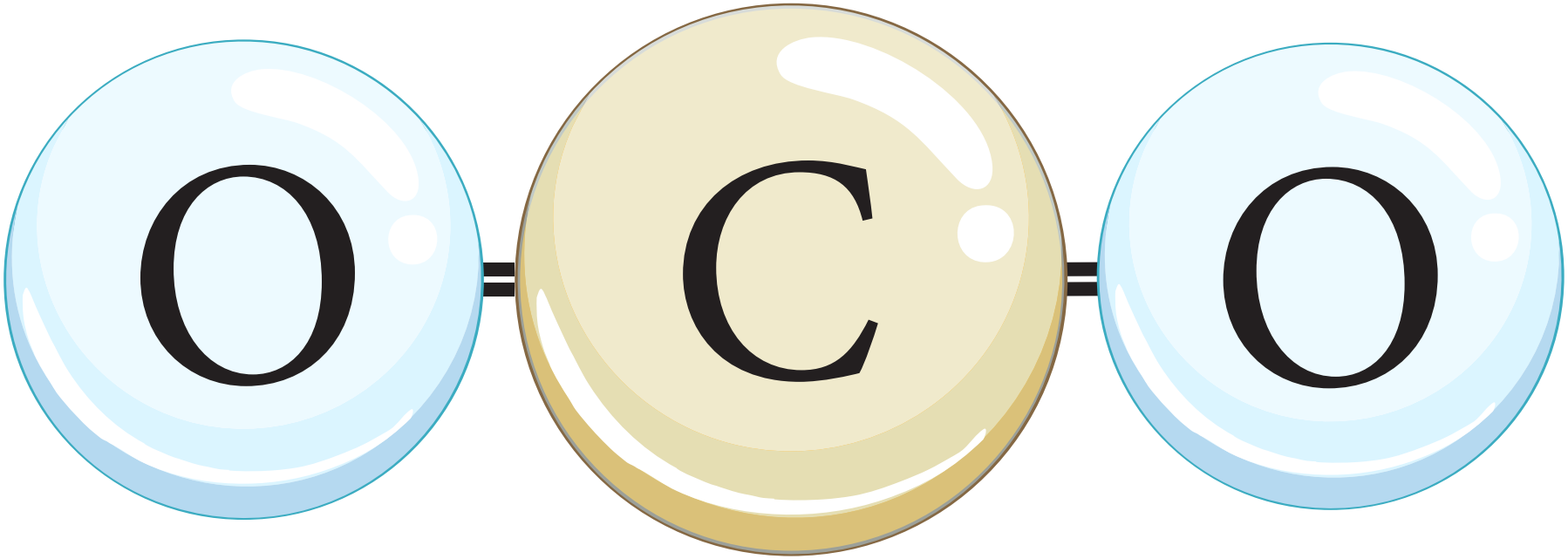
- Ozone in the Troposphere
- Ozone in Our Neighborhood
- EPA: Ground-level Ozone Basics

NATIONAL OZONE GARDEN NETWORK

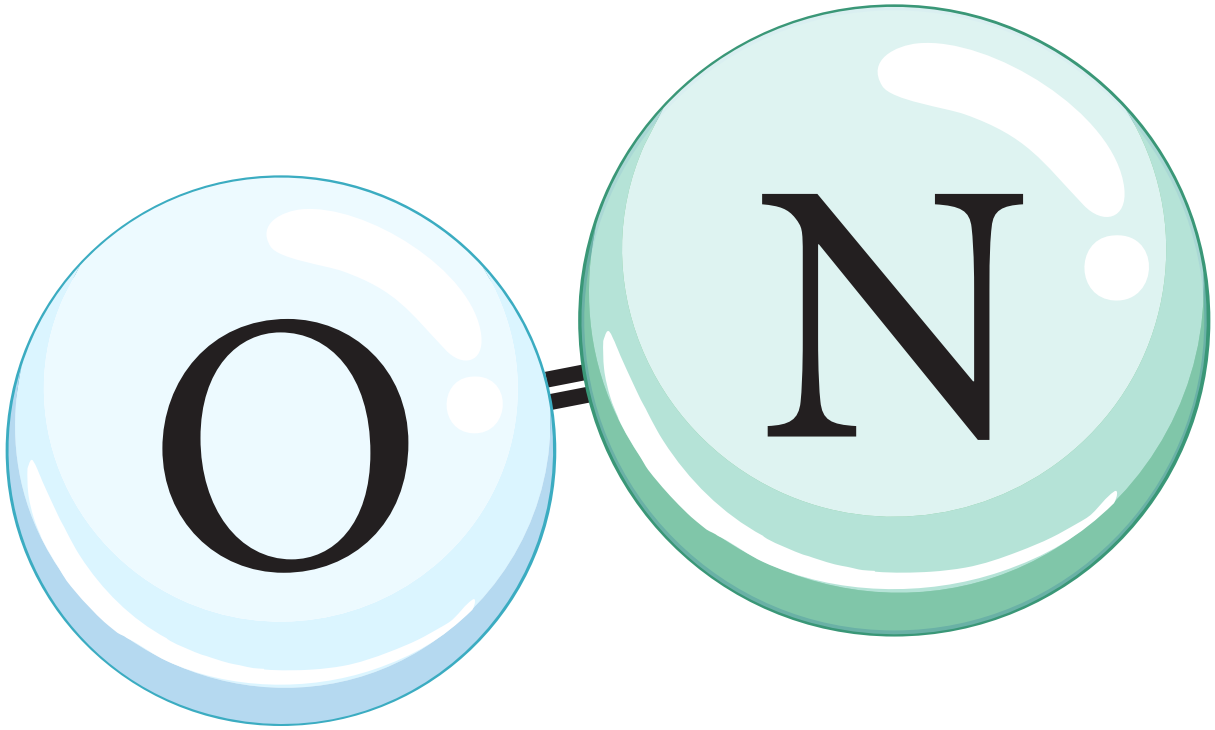
UCAR CENTER FOR SCIENCE EDUCATION

© 2025 University Corporation for Atmospheric Research *All rights reserved*

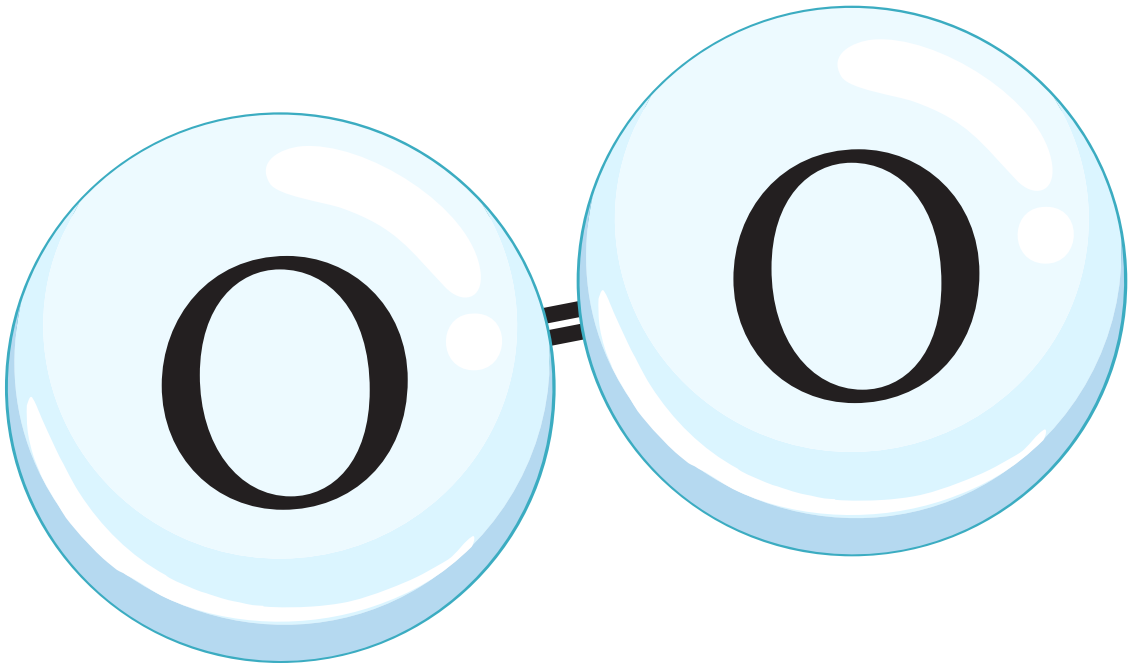




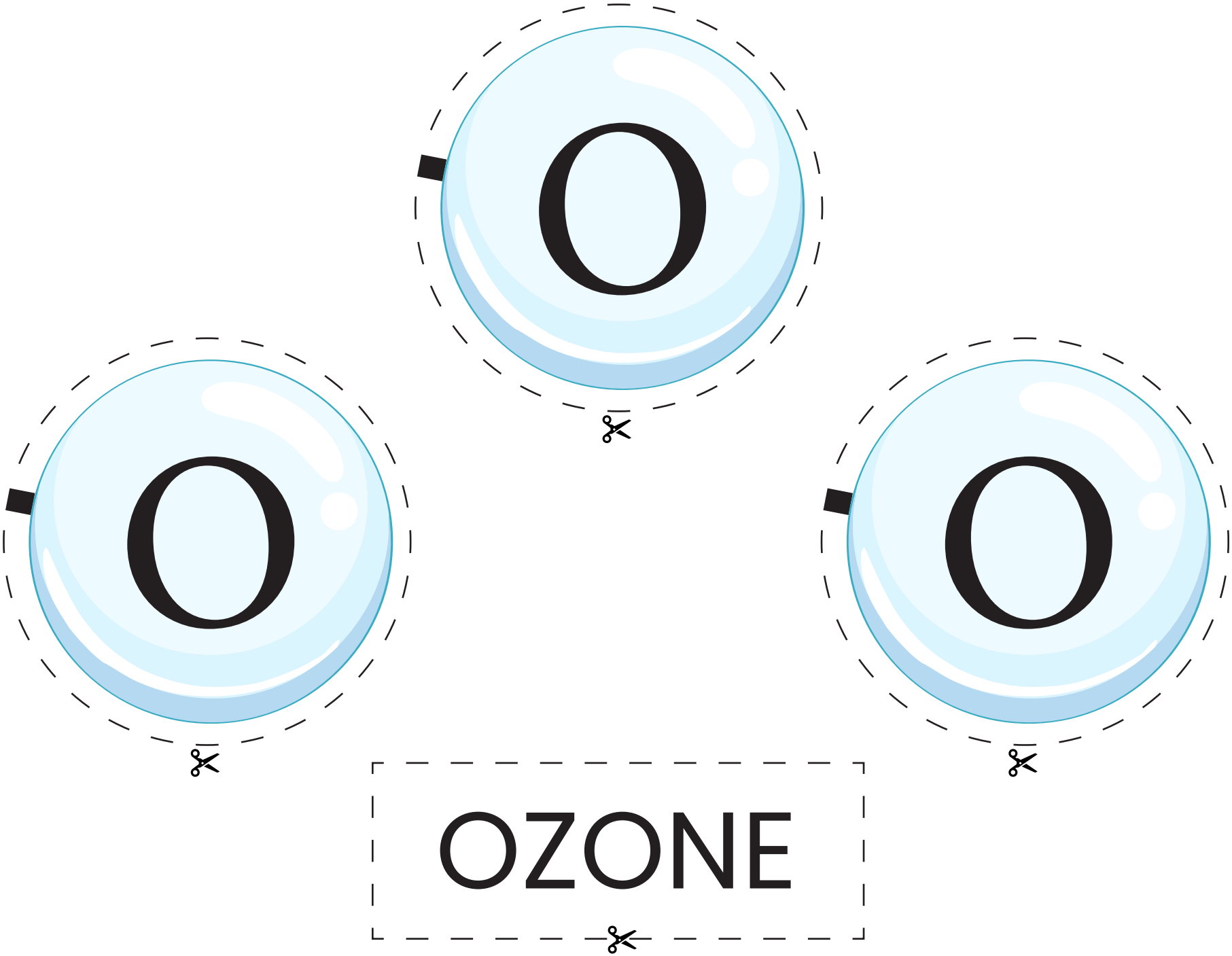
CO_2

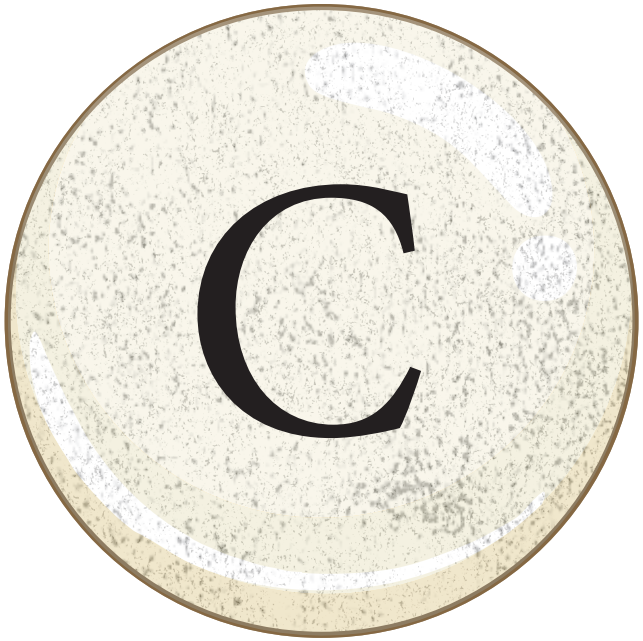


NO_2



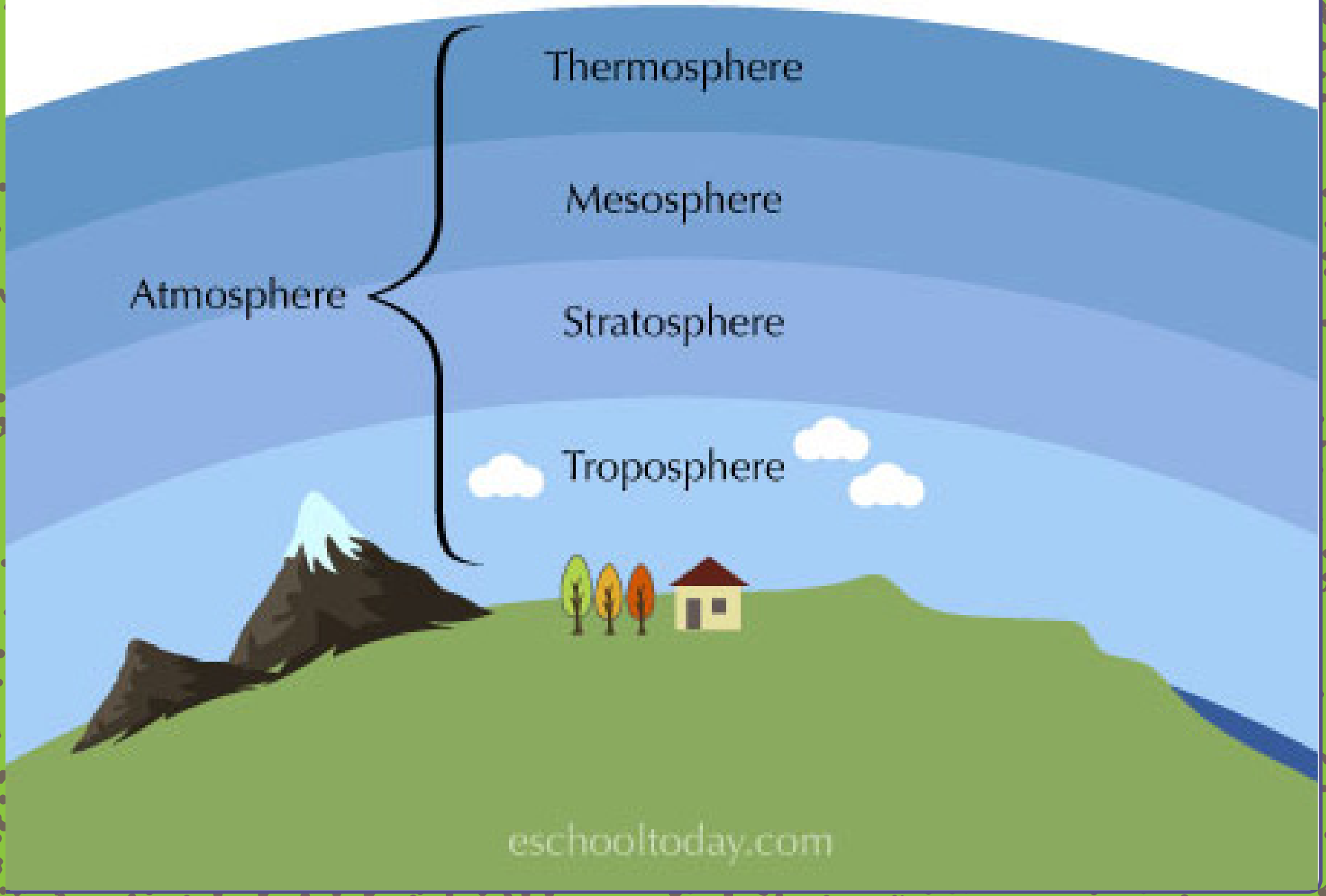
O_2

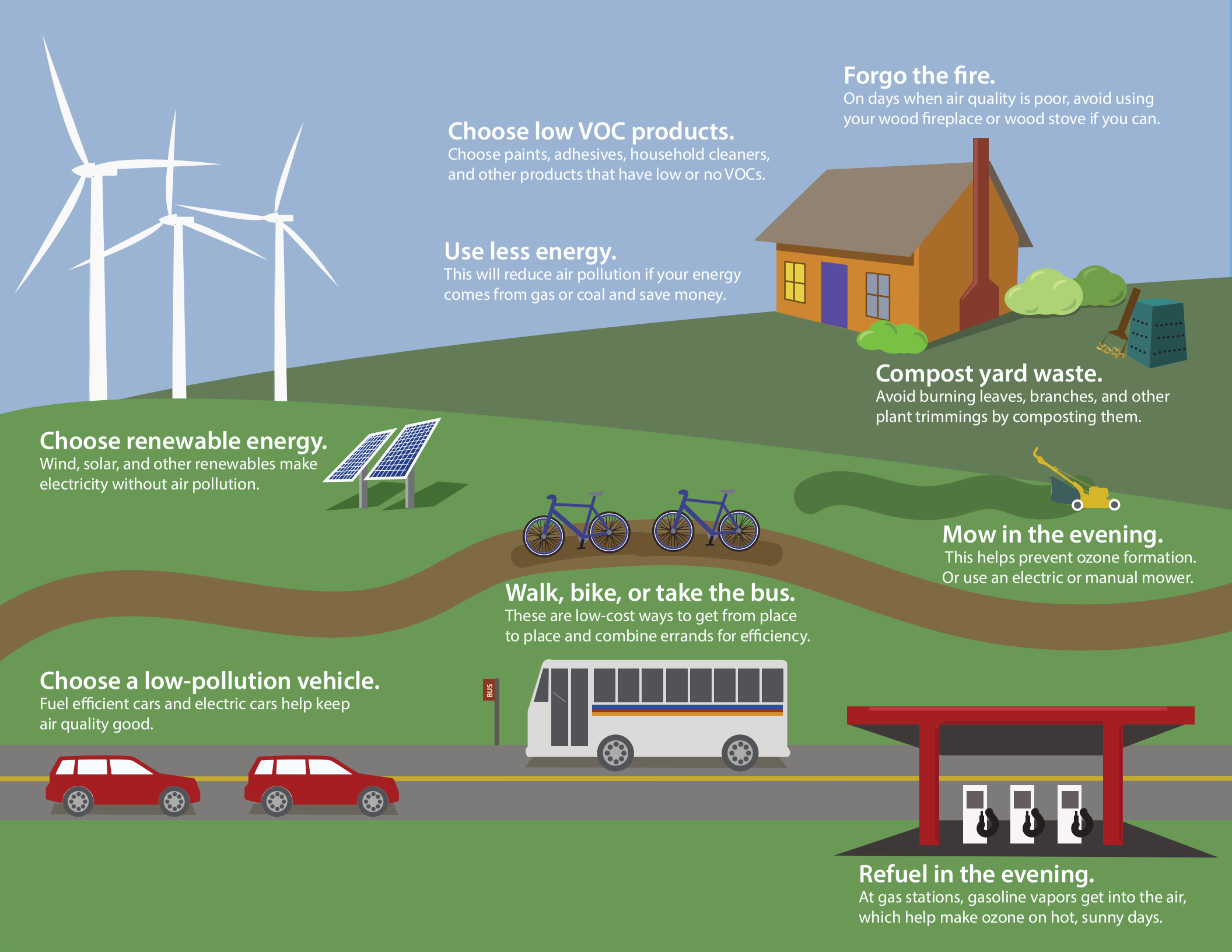

O
O
O
OZONE



Carbon

VOC


Thermosphere
Mesosphere
Atmosphere
Stratosphere
Troposphere
eschooltoday.com


Forgo the fire.
On days when air quality is poor, avoid using your wood fireplace or wood stove if you can.
Choose low VOC products.
Choose paints, adhesives, household cleaners, and other products that have low or no VOCs.
Use less energy.
This will reduce air pollution if your energy comes from gas or coal and save money.
Compost yard waste.
Avoid burning leaves, branches, and other plant trimmings by composting them.
Choose renewable energy.
Wind, solar, and other renewables make electricity without air pollution.
Mow in the evening.
This helps prevent ozone formation. Or use an electric or manual mower.
Walk, bike, or take the bus.
These are low-cost ways to get from place to place and combine errands for efficiency.
Choose a low-pollution vehicle.
Fuel efficient cars and electric cars help keep air quality good.
BUS
Refuel in the evening.
At gas stations, gasoline vapors get into the air, which help make ozone on hot, sunny days.